# torch.nn.functional.softmax#

https://pytorch.org/docs/stable/generated/torch.nn.functional.softmax.html

**Description:**

Apply a softmax function. Softmax is defined as: Softmax(xi)=exp(xi)∑jexp(xj)\text{Softmax}(x_{i}) = \frac{\exp(x_i)}{\sum_j \exp(x_j)}Softmax(xi)=∑jexp(xj)exp(xi) It is applied to all slices along dim, and will re-scale them so that the elements lie in the range [0, 1] and sum to 1. See Softmax for more details.

## Function Signature

```
torch.nn.functional.softmax(input, dim=None, _stacklevel=3, dtype=None)[source]#

Parameters

Return type
```

**Returns:**

Tensor

## Notes & Warnings

**NOTE**

Note This function doesn't work directly with NLLLoss, which expects the Log to be computed between the Softmax and itself. Use log_softmax instead (it's faster and has better numerical properties).

# Detailed Documentation

## Documentation

Apply a softmax function.

Softmax is defined as:

Softmax(xi)=exp(xi)∑jexp(xj)\text{Softmax}(x_{i}) = \frac{\exp(x_i)}{\sum_j \exp(x_j)}Softmax(xi)=∑jexp(xj)exp(xi)

It is applied to all slices along dim, and will re-scale them so that the elements lie in the range [0, 1] and sum to 1.

See Softmax for more details.

input (Tensor) – input

dim (int) – A dimension along which softmax will be computed.

dtype (torch.dtype, optional) – the desired data type of returned tensor. If specified, the input tensor is casted to dtype before the operation is performed. This is useful for preventing data type overflows. Default: None.

This function doesn’t work directly with NLLLoss, which expects the Log to be computed between the Softmax and itself. Use log_softmax instead (it’s faster and has better numerical properties).